

Final Dataset Validation Report

1. Ward_ID / Ward_Name consistency across all datasets

- ahmedabad_wards_clean.geojson: 48 wards, matches geojson exactly: 
- demographic_join_template.csv: 48 wards, matches geojson exactly: 
- ward_thermal_daily.csv: 48 wards, matches geojson exactly: 
- priority_rubric_v2_full_test.csv: 48 wards, matches geojson exactly: 
- action_engine_full_test.csv: 48 wards, matches geojson exactly: 

2. Missing values

- ward_thermal_daily.csv: 0 missing values (expected 0)
- demographic_join_template.csv real columns (population_total, population_density, ndvi_green_cover): 0 missing (expected 0)
- demographic_join_template.csv pending columns (elderly_pct etc.): 288 / 288 null — EXPECTED, documented in Decision Log D7/D8 as not yet available (not a data-quality bug)
- priority_rubric_v2_full_test.csv core columns: 0 missing (expected 0)
- action_engine_full_test.csv: 0 missing tier/action values (expected 0)

3. Unit and range sanity checks

- UTCI_mean stays within a physically plausible range (-10 to 60°C): 
- NCTL is never negative (by definition, max(0,·) sum): 
- CTBI stays within [0,1] (per D6/D6a normalization audit): 
- population_density within a plausible Ahmedabad range (500-60,000/km²): 
- ndvi_green_cover within valid NDVI range [-1,1]: 
- Total estimated population across 48 wards: 6,094,460 (Ahmedabad's actual ~6.3-6.5 million —  plausible)

4. Dashboard (src/app.py) numbers vs latest computed CSVs

- Dashboard currently loads: `priority_rubric_v1_full_test.csv`
-  **FINDING (real, not cosmetic):** The dashboard is still reading **v1** priority data (thermal-only, no vulnerability). Since then, the team has built:
 - `priority_rubric_v2_full_test.csv` (D10) — real vulnerability (population density + NDVI) integrated
 - `action_engine_full_test.csv` (Day 4) — real, HAP-grounded action recommendations per ward

The dashboard's sidebar text still says *"vulnerability & response capacity data are still not available (manual AMC download still needed)"* — **this is now stale**. Vulnerability data has been available since D8/D10. The dashboard needs to be pointed at `priority_rubric_v2_full_test.csv` and `action_engine_full_test.csv` instead, or a judge who reads the sidebar will be told something false about the project's own progress.

- v1 (currently shown on dashboard) tier split, latest date: {'High': 48}
- v2 (actual latest science) tier split, same date: {'MODERATE': 25, 'HIGH': 23}
- These are meaningfully different — v1 shows only thermal (mostly one tier city-wide, since CTBI is a single city-wide series), while v2 shows REAL ward-to-ward variation (23 High / 25 Moderate) because vulnerability now differs per ward.

Showing v1 in a live demo would understate the project's actual progress and hide the spatial variation the team specifically worked to add.

Overall verdict: **ACTION NEEDED** — see Section 4

Data quality itself (Sections 1-3) is clean — no missing values, no ward mismatches, all ranges physically sensible. The one real issue is integration lag: the dashboard hasn't caught up to the team's latest outputs (v2 + Action Engine). This is a wiring fix (change two file paths + a few column names in `app.py`), not a data problem — flagged for Vrinda to pick up next.